Narrative Consistency Classification over Long-Form Text

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1. Overall Approach

This particular problem will attempt to identify if the character's story has been logically consistent in the entire long story. The most challenging aspect of the problem statement is that the problem involves analysing long narratives that are in excess of 100,000 words, where the important details in the story are presented in text that is spread out and cannot be viewed through the local perspective.
In order to deal with this, it introduces a new approach on how to treat narrative consistency check as a classification problem from a retrieval perspective, rather than addressing it from a generative perspective. The important aspect of this introduction is breaking away from connecting long-context understanding with reasoning.

1.1 Problem Framing as a Decision Task

In place of asking the language model to explain the summary of the novel, the language model is given a choice among two alternatives which has to be purely binary:

- Can there be a possibility that the suggested background might be able to account for the events, the character traits, and the resulting events observable within the novel?
- "Does it honour the implicit narrative constraints of character motivations, obligations, and time?"

This is appropriate to the challenge's requirement of "global consistency," which helps to offset troubles with surface-level plausibility.

1.2 Retrieval-Augmented Reasoning Pipeline

Basically, the system works in a four-step process:

1. Indexing of Long-Documents:
   Each novel is fully semantically chunked into fixed-size pieces. These chunks are then embedded and stored in a vector similarity database, allowing the access to any part of the narrative semantically without truncation.
2. Evidence Retrieval:
   The model retrieves top-K most relevant narrative passages for a given backstory. Retrieval queries combine the details of the backstory and metadata of the novel to capture both character-specific and contextual signals.
3. Constraint-Oriented Deduction:
   The model goes through retrieved passages, while a carefully designed prompt instructs it to perform the following tasks:
   - Assess consistency over time
   - Assess causal plausibility
   - Identify violations of implicit narrative constraints
   - Base conclusions on combined evidence rather than isolated passages
4. Output of Binary Classification:
   The model is limited to generating a clear output that includes the following:

- A binary consistency label (0 or 1)
- Short evidence-based explanation

This design ensures that reasoning is bound, clear, and verifiable.

1.3 Robustness-Oriented System Design

Since LLM models are known to demonstrate instability, this model uses several methods of achieving robustness:

- Several approaches to using the LLM to facilitate fallback where there is API/model failure
- A response validation and exclusion of invalid and completed responses
- Retry mechanism incorporating a heuristic fallback for the purpose of ensuring predictability of results across runs

These allow the result to be reproduced and are not dependent on one weak chain of inference.

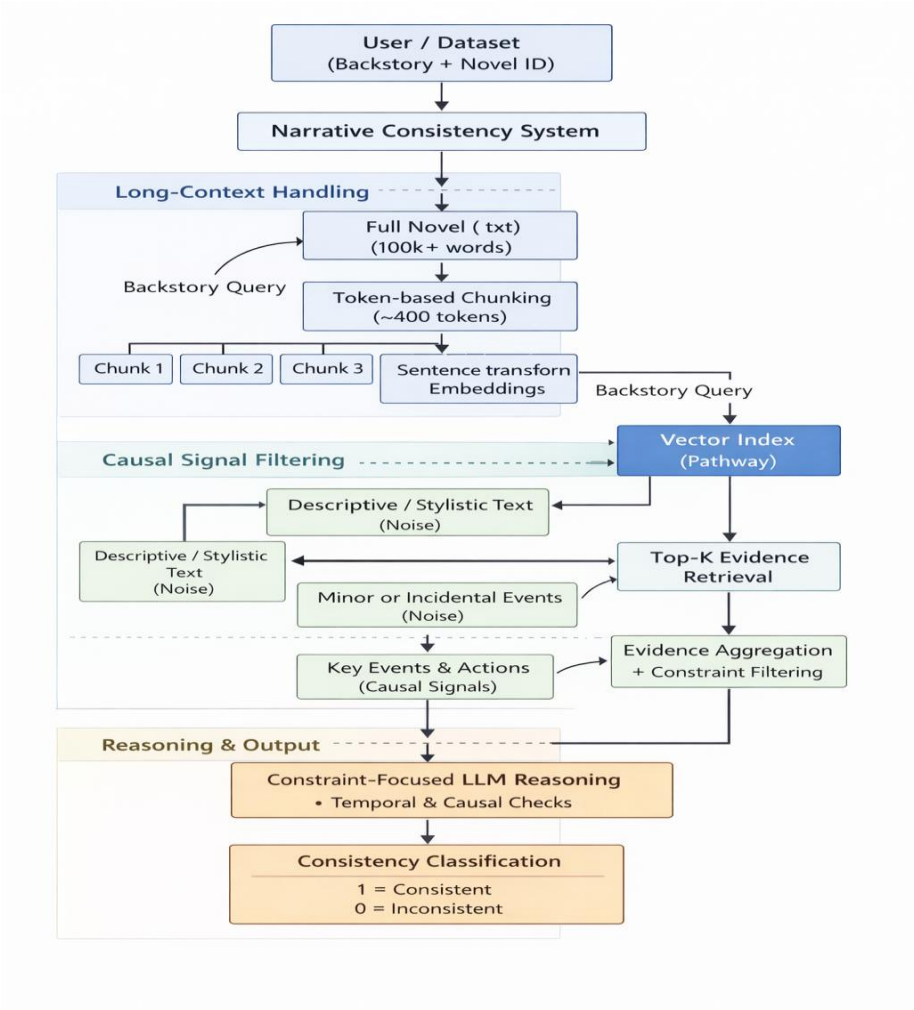

Figure 1: Overall Architecture

1.4 Key Design Principles

The whole approach is based on the following principles:

- Evidence rather than plausibility: A decision should be made only on the basis of the narrative evidence which has been
- Selective context vs. full context: Retrieval is better than full novel prompt generation.
- Causal fit versus similarity: Logic consistency is favoured over similarity.
- Resistance and Model Complexity: Resistance to engineering rather than pursuing complex models.

2. Handling Long Context

However, it is important for it to also be able to carry out high-quality reasoning on these MMs. Since each novel has more than 100,000 words, and most of the information that will be used for inferencing will be dispersed across different parts of the story, it will not be feasible for it to be prompted with the entire text for it to be able to see everything at once that will guide its responses.
Specifically, in terms of addressing this issue, it implements a selective retrieval approach and gathering of evidence such that it is globally consistent but computationally restricted.

2.1 Full Document Ingestion without

All novels are fully processed, with no filtration or summarization. Books are handled by the computer program as full sources of information, rather than partial information. All the above ensures that:

- Long range dependencies are available
- Early story constraints may influence subsequent reasoning
- There are no assumptions in lossy preprocessing

Consuming the whole file satisfies the challenge's requirement for reasoning over entire stories instead of extracts.

2.2 Token-Based Semantic Chunk

Since processing raw text files for long documents proves to be not so practical, each novel is split using a token count-based chunking technique:

- There is a maximum number of tokens for chunks.
- The size of a chunk is determined in order to take into account the semantic consistency and the details of the retrieval process
- Chunking retains local story structure. In this case, it involves events, dialogues, and actions.

This ensures that each chunk is still significant independently while facilitating the indexing and retrieval process on the entire novel.

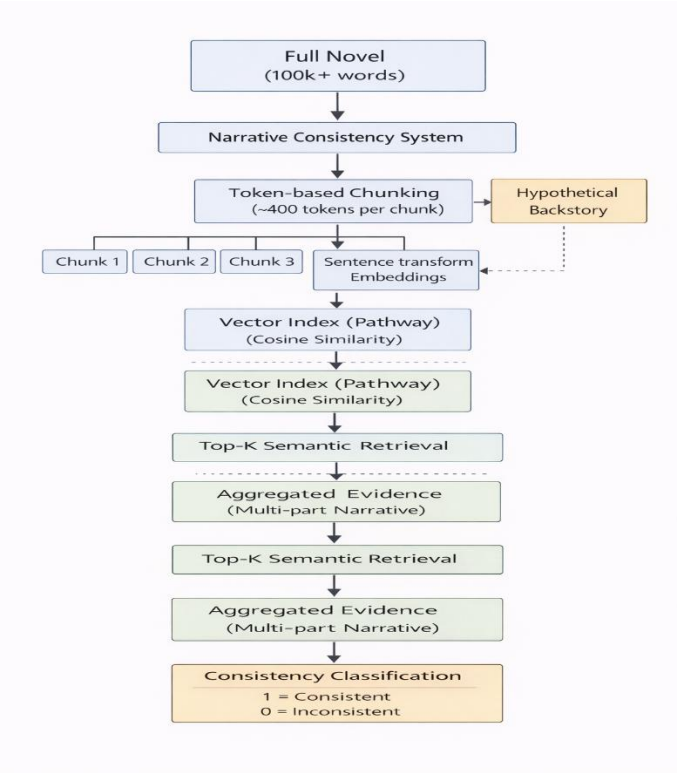

Figure 2: Long Context Handling Flowchart

2.3 Dense Vector Representation and Indexing

Every chunk is embedded using a sentence-level transformer model, which produces a set of dense semantic representations. These representations are stored in a vector similarity index, which is capable of supporting efficient nearest-neighbour queries.

The dense representation helps the system in the following ways:

- Extract quotes with semantic relevance rather than keywords
- Identify paraphrased/related narrative events
- Weave together elements of the backstory into distant but relevant story sections

The significance of this embedding strategy also lies in helping to cope with the sparsity and indirection that exists in narrative constraints.

2.4 Evidence-Oriented

During the inference phase, the model is able to fetch a small fixed number of top-scoring candidates for each query in the backstory. These queries consist of the combination of:

- “Backstory” details
- Novel metadata (e.g., Title)

This will help trigger the recall of passages related to the development of the characters, the key occurrences in the plot, and the resolutions.

2.5 Aggregation Across Multiple Narrative Locations

Rather than relying on a single retrieved passage, the system aggregates evidence across multiple chunks drawn from different parts of the novel. This design choice is essential for:

- Capturing character evolution over time
- Detecting contradictions between early and late narrative states
- Avoiding false positives caused by isolated supporting evidence

Aggregated evidence allows the model to reason over patterns and constraints, not just individual facts.

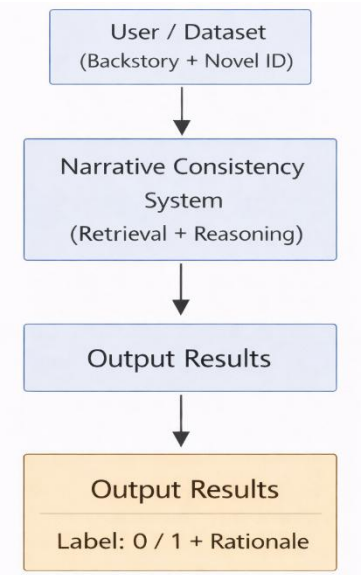

Figure 3: Narrative Refine Process

2.6 Secondary Evidence Recovery Mechanism

To enhance longer context resistance further, a final way of extracting fallback evidence is employed when the quality of vector retrieval is indefinite. This involves: Diagram 3:

- Picks out key keywords from the history
- Ranks narrative paragraphs according to overlapping keywords
- Chooses the highest-rated passages as additional evidence.

This ensures that the system remains robust even when embedding-based retrieval fails to yield critical context.

2.7 Separation of Retrieval and Reasoning

An important design point is to distinguish sharply between retrieval and reasoning:

- Retrieval handles size and extent in long stories
- Reasoning relies solely on stipulated high-quality evidence

This division of labor helps to reduce the cognitive load on the language model as well as improve global consistency properties since reasoning would be performed on information relevant to constraints rather than on the volume of stories.

2.8 Benefits for Global Coherence

The strategy to deal with long contexts is beneficial in a number of ways:

- Facilitates reasoning between remote parts in a narrative
- "Preserves causal links across multiple chapters"
- Minimizes the risk of biased narratives locally
- Scales well with the length of the story

On the whole, it ensures global story-line cohesion without requiring full context generation or memory-intensive architectures.

3. Distinguishing Causal Signals from Noise

In a number of long-story texts, there are many cases in which text fragments could seem to semantically connect with the backstory, although they do not explicitly state strong connections of causality. To resist the comparison based on superficial similarities, the model demonstrates preference for causal and temporal connections over similarities in the text.

The evidence is gleaned from various places within the text and constructed before starting the reasoning process, so there is less reliance on isolated pieces of evidence that could be a coincidental fit or not even be relevant. The reasoning problem helps the machine learn if the hypothesized history could possibly lead to the observed later actions of the characters.

It is, for example, observed that implicit constraints in narrative, such as the kinds of character motivations and commitments, have been seen to be recognized as genuine sources of inconsistency without the requirement of conflict at the sentence level. In this light, such strategies are in tandem with the analysis of robustness in outputs in efforts to ensure the predictions are underpinned by meaningful causality rather than narrative chatter.

4. Robust Inference and Reproducibility

For ensuring a completely reproducible execution, the system provides:

- Pathway document ingestion and retrieval
- Deterministic prompt format and output structure
- Strong fallback strategies for LLM:
    - Multiple methods for accessing LLM
    - Retry Logic
    - Response validation
    - Safe alternatives to bad outputs

All predictions are made fully automated from the input variables without the need for any human interventions, thereby satisfying the criterion of being reproducible.

5. Results File Generation

Predictions are written to a CSV file named **results.csv**, with one row per test example:

id	prediction	rationale
…	0/1	Concise Explanation

The system also performs basic sanity checks to detect degenerate outputs (e.g., all predictions being identical).

6. Limitations and Failure Cases

A large number of the system's runs depend on the quality of the evidence retrieved. Although the method of semantic evidence retrieval is highly successful in discovering appropriate passages in the text, it fails to notice the implicit restrictions on the narrative sometimes, especially in the event of gradual character growth. The final consistency judgment is based on LLM reasoning. While strict prompts and robustness tests can ensure a minimal number of errors, there could still be issues concerning indirect psychological inconsistencies or distant causal connections. Such limitations lie within the state of the art of model reasoning and not in any issues of system architecture.

Secondly, this model is relatively less sensitive to thematic inconsistencies rather than actions. The fall-back strategies will always guarantee stable predictions in case of inference failure, but thematic inconsistencies might be dealt with less effectively by these simplified strategies.

7. Conclusion

The significance of the proposed project is that it develops a strong approach to evaluating the consistency of conjectural character biographies with long pieces of text through the approach of making a binary decision using retrieved information, which prevents the weak approach involved in generating text in full context.

With these capabilities involving semantic chunking, vector-based retrievals, and constraint-oriented prompting, it is able to facilitate strong narrative reasoning past context size limitations. Although it lacks the ability to track dynamic character states or symbolic story structures, it proves that combining evidence aggregation with controlled LLM-based reasoning is an efficient method for long context-based narrative consistency classification.